

Defect modes: the symmetric-power wall as spectral synthesis on the denominator's zeros

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Abstract

Let f be a level-one Hecke eigenform, T_5, T_3 the one-sided theta functions of the Sym^5 and Sym^3 banks with their completed archimedean kernels, and $T_P = T_5 \star T_3$ (multiplicative convolution) the theta of the Rankin–Selberg pair $f \times \text{Sym}^4 f$. The pair and the denominator are automorphic (Jacquet–Piatetski-Shapiro–Shalika; Kim–Shahidi), so their reflections are theorems; the Sym^5 reflection is the open object. We prove an exact identity: the pair's reflection defect equals the T_3 -convolution of the Sym^5 defect, so the Sym^5 defect D_5 is annihilated by every multiplicative translate of the cuspidal theta T_3 — unconditionally. Because T_3 is exponentially small at both ends of the carrier, its Mellin transform is the entire function $\Lambda(\text{Sym}^3)$, and the annihilation system admits exactly the paired zero-mode solutions $v^\rho - \varepsilon_5 v^{1-\rho}$, $\rho \in Z(\Lambda(\text{Sym}^3))$. The containment statement (entirety of $\Lambda(\text{Sym}^5)$, hence the whole symmetric-power ladder past the classical record) is therefore precisely a spectral-synthesis statement: the actual defect, a priori a superposition of denominator-zero modes, has no modes. The same identity holds for every Dirichlet twist, pinning each twisted defect to the corresponding twisted zero set. Soft tauberian arguments provably cannot close this — the modes exist — and the remaining content is the arithmetic coherence of the mode coefficients across the twisted family.

1 Setup and the identity

Fix the level-one eigenform f of weight k (the numerics elsewhere use $k = 12$), analytic normalization throughout. For a rank- d symmetric-power bank with coefficients $\lambda(n)$ and archimedean kernel K (the inverse Mellin of its $\Gamma_{\mathbb{R}}$ -stack), the one-sided theta is $T(x) = \sum_{n \geq 1} \lambda(n) K(nx)$, absolutely convergent on $(0, \infty)$, and its reflection defect is

$$D(x) = T(1/x) - \varepsilon x T(x).$$

The reflection statement “ $D \equiv 0$ ” is equivalent, through the mechanism ring, to the full one-sided analytic package of the bank. Write T_5, T_3 for the $\text{Sym}^5, \text{Sym}^3$ thetas, $\varepsilon_5, \varepsilon_3$ their signs (both -1 for $k = 12$, measured and consistent with the classical local computations), and

$$T_P(x) = (T_5 \star T_3)(x) = \int_0^\infty T_5(x/t) T_3(t) \frac{dv}{v} [t]$$

*Companion to *The Sato–Tate inputs for a Maass form* and *The information capacity of functional-equation windows*, same repository. sam@sk-labs.com

the theta of the pair $f \times \text{Sym}^4 f$: its kernel is the product stack and its coefficients the Dirichlet convolution, so this is the compiled bank identity $\text{std} \otimes \text{Sym}^4 = \text{Sym}^5 \oplus \text{Sym}^3$ at function level; the integral converges absolutely (each theta decays faster than any power at ∞ and grows at most polynomially at 0).

Theorem 1.1 (defect convolution). *With D_P the pair defect and $D_3 \equiv 0$ the denominator's (Kim–Shahidi),*

$$D_P(x) = -\varepsilon_5 \varepsilon_3 x \int_0^\infty D_5(v) T_3(xv) \frac{dv}{v}.$$

In particular, since the pair reflects (Jacquet–Piatetski-Shapiro–Shalika), $D_P \equiv 0$ and

$$\int_0^\infty D_5(v) T_3(xv) \frac{dv}{v} = 0 \quad \text{for every } x > 0.$$

Proof. In $D_P(x) = T_P(1/x) - \varepsilon_5 \varepsilon_3 x T_P(x)$ substitute $t \mapsto 1/(xw)$ in the first integral: $T_P(1/x) = \int T_5(w) T_3(1/(xw)) \frac{dw}{w}$, and insert the denominator's reflection $T_3(1/(xw)) = \varepsilon_3 xw T_3(xw)$ (exact, $D_3 \equiv 0$). In the second integral substitute $t = xv$: $x T_P(x) = x \int T_5(1/v) T_3(xv) \frac{dv}{v}$, and insert the definition $T_5(1/v) = \varepsilon_5 v T_5(v) + D_5(v)$. The two $\int v T_5(v) T_3(xv) dv$ terms cancel and the displayed identity remains. All integrals converge absolutely: at $t \rightarrow \infty$ the theta factors decay super-polynomially, at $t \rightarrow 0$ the partner factor does; splitting D_5 into its two terms keeps absolute convergence since $T_5(1/v)$ is small precisely where $T_3(xv)$ grows. Fubini and the substitutions are then legitimate. $\square$

2 The mode space

Proposition 2.1 (cuspidal two-sided decay). *T_3 is exponentially small at both ends: at ∞ by kernel decay, at 0 by its own reflection, $T_3(y) = \varepsilon_3 y^{-1} T_3(1/y)$. Hence its Mellin transform converges for every $s \in \mathbb{C}$ and equals the entire completed function $\Lambda(\text{Sym}^3, s)$.*

Theorem 2.2 (zero modes). *For every ρ with $\Lambda(\text{Sym}^3, \rho) = 0$ the function*

$$D_\rho(v) = v^\rho - \varepsilon_5 v^{1-\rho}$$

satisfies both constraints of the defect system: the annihilation $\int_0^\infty D_\rho(v) T_3(xv) \frac{dv}{v} = x^{-\rho} \Lambda(\text{Sym}^3, \rho) - \varepsilon_5 x^{\rho-1} \Lambda(\text{Sym}^3, 1-\rho) = 0$ (the functional equation pairs ρ with $1-\rho$), and the forced anti-reflection $D(1/v) = -\varepsilon_5 v^{-1} D(v)$ that every reflection defect obeys. The system therefore has nonzero solutions of exactly the defect's growth class, one (with multiplicity) for each zero of the denominator's completed function.

Corollary 2.3 (the wall, exactly). *No argument using only the pair's and the denominator's reflections and growth — in particular no Wiener–Beurling tauberian argument — can force $D_5 \equiv 0$: the solution space is as large as $Z(\Lambda(\text{Sym}^3))$. Entirety of $\Lambda(\text{Sym}^5)$ (equivalently the zero-containment $Z(\Lambda(\text{Sym}^3)) \subseteq Z(\Lambda(f \times \text{Sym}^4 f))$, equivalently the vanishing of every mode coefficient of the actual defect D_5) is a spectral synthesis statement on the denominator's zero set.*

Remark 2.4. This is, we believe, the sharpest available form of the symmetric-power wall: not “an L -function lacks continuation” but “a concrete function, annihilated by a concrete translate family, must have empty mode spectrum.” The failure of the soft route is now a theorem rather than an experience.

3 The twisted family and the endgame

For every primitive Dirichlet character χ the pair $(f \otimes \chi) \times \text{Sym}^4 f$ is again an automorphic Rankin–Selberg pair and $\text{Sym}^3 f \otimes \chi$ is again Kim–Shahidi, so Theorem 1.1 holds verbatim with twisted data: the twisted defect D_5^χ is annihilated by the translates of $T_{3,\chi}$ and is therefore a superposition of $Z(\Lambda(\text{Sym}^3 \otimes \chi))$ -modes. Unconditionally, then, the family $\{D_5^\chi\}_\chi$ is a system of mode-superpositions over the twisted zero sets, all generated by the *single* coefficient sequence $\lambda_5(n)$. The endgame is the arithmetic coherence of this system: mode coefficients are Mellin residues of the one meromorphic continuation $\Lambda(\text{Sym}^5 \otimes \chi)$, the twisted zero sets vary with χ while the Euler product does not, and the question — the last question — is whether any nonzero Hecke-multiplicative sequence can feed a coherent nonempty mode system. This is the Weil-converse shape with the twist information localized on zero sets rather than spread over strips; our capacity measurements (companion note) quantify why the $\text{GL}(1)$ family alone carries limited depth, and the mode formulation is where higher-rank twists’ light channels would act.

Remark 3.1 (register). Theorem 1.1 and 2.2 are unconditional given the cited automorphy of the pair and the denominator; no claim is made that the mode space exhausts the solution class beyond the exhibited span (spectral synthesis may admit closures), and none is needed: the exhibited modes already defeat the soft route, and the synthesis formulation of containment stands independently. The numerical containment check (repository, `tmp/containment_check.py`) tests the statement zero-by-zero at the first zeros of $\Lambda(\text{Sym}^3 \Delta)$.

References

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